

Vishnu Vardan Babu Pentela

Full Stack Python Developer

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PROFESSIONAL SUMMARY:

3+ years of experience as a Full Stack Python Developer specializing in architecting, developing and deploying scalable, secure and high-performance applications. Expert in crafting robust backend microservices with FastAPI, Django REST Framework and SQLAlchemy, integrating complex financial (QuantLib, Plaid, Stripe, PayPal) and healthcare (FHIR, HL7, EDI 837/835) APIs and optimizing data flow with TimescaleDB, Redis and Celery. Proficient in building dynamic, real-time UIs using React.js, Vue.js, Tailwind CSS, Plotly, Dash and WebSockets. Extensive experience with secure authentication (OAuth 2.0, JWT, OpenID Connect, RBAC), compliance (HIPAA, GDPR, ISO 27001, Indian IT Act 2000, AES-256) and automating CI/CD pipelines via GitHub Actions, Terraform and Ansible for multi-region AWS deployments (EC2, S3, RDS, Lambda, Kubernetes, Docker). Proven ability in comprehensive testing (PyTest, Selenium, Postman) to deliver resilient, error-free solutions.

TECHNICAL SKILLS:

Languages & Frameworks:	Python (FastAPI, Django, Flask, SQLAlchemy), JavaScript (React.js, Vue.js), TypeScript, Tailwind CSS, HTML5, CSS3
API Design & Integration:	RESTful APIs, GraphQL, OAuth 2.0, JWT, OpenID Connect, Plaid, Stripe, PayPal, EDI 837/835, HL7, FHIR
Frontend Development:	React.js, Vue.js, Next.js, Dash, Plotly, WebSocket
Backend & Microservices:	FastAPI, Django REST Framework, Celery, Redis, RabbitMQ
Database Technologies:	PostgreSQL, TimescaleDB, MySQL, AWS RDS, NoSQL, SQL
Cloud & DevOps:	AWS (EC2, S3, RDS, Lambda), Docker, Kubernetes, GitHub Actions, Terraform, Ansible, IaC
CI/CD & Testing:	Git, GitHub, PyTest, Selenium, Postman, Jenkins
Security & Compliance:	RBAC, AES-256, HIPAA, GDPR, ISO 27001, Indian IT Act 2000
Analytics & Visualization:	Pandas, NumPy, Apache Superset, Dash, Plotly
Tools & Platforms:	Jira, Linux, Postman, Figma, Swagger, ELK Stack, Prometheus, Grafana

PROFESSIONAL EXPERIENCE:

KPMG US – NJ | Python Developer | September 2024 – Present

- Collaborated with financial analysts and product owners to translate complex investment strategies into backend logic using QuantLib, ensuring alignment with trading objectives and regulatory requirements.
- Developed asynchronous RESTful microservices using FastAPI and SQLAlchemy, implementing core trading and risk management logic and deployed them as Docker containers on Kubernetes clusters hosted on AWS EC2.
- Built secure user authentication and authorization workflows using OAuth 2.0, JWT and OpenID Connect, enabling Role-Based Access Control (RBAC) and ensuring data privacy and secure access across user tiers.
- Integrated third-party financial APIs including Plaid, Stripe and PayPal SDKs for seamless user onboarding, account aggregation, payment processing and transaction categorization.
- Engineered time-series database architecture with TimescaleDB to efficiently store and query high-frequency trade and market data, powering analytics and historical performance dashboards.
- Designed and developed a real-time trading dashboard with React.js, leveraging WebSockets to stream live price updates, trade confirmations and portfolio metrics with sub-second latency.
- Reduced transaction processing latency by 68% and improved system resilience by implementing asynchronous event queues with Redis and Celery for trade execution and notification workflows.
- Built CI/CD pipelines using GitHub Actions and Terraform for infrastructure-as-code (IaC), enabling automated testing, secure deployments and zero-downtime releases across multi-region environments.

Deloitte – India | Python Developer | April 2021 - July 2023

- Developed scalable RESTful and GraphQL APIs using Django REST Framework and FastAPI to enable seamless integration with Electronic Health Record (EHR) & Electronic Medical Record (EMR) systems, improving data accessibility for healthcare providers by 20%.
- Implemented FHIR and HL7 standards to ensure interoperable patient data exchange across platforms like EPIC, Cerner, & Indian EHR systems (Ayushman Bharat Digital Mission), enhancing care coordination across multi-specialty hospitals.
- Developed responsive telemedicine interfaces using Vue.js, React and Tailwind CSS, ensuring WCAG 2.1-compliant accessibility for diverse patient demographics, including rural and semi-urban populations in India.
- Built real-time patient monitoring dashboards leveraging Plotly, Dash and WebSocket for visualizing critical vitals and clinical data, enabling healthcare providers to reduce response times by 30% in critical care scenarios.
- Integrated clinical analytics solutions using Pandas, NumPy and Apache Superset to create data-driven dashboards, empowering providers with actionable insights for 15% faster clinical decision-making.
- Ensured compliance with HIPAA, GDPR and Indian IT Act 2000 by implementing OAuth 2.0, JWT-based authentication and AES-256 encryption for secure patient data transmission and storage across cloud-based healthcare applications.
- Automated medical billing and claims processing by integrating EDI 837/835 standards with Indian healthcare payer systems, reducing claim processing errors by 25% and improving reimbursement cycles.
- Implemented Role-Based Access Control (RBAC) and fine-grained access policies aligned with ISO 27001 and Indian cybersecurity guidelines, safeguarding sensitive patient data in multi-tenant healthcare platforms.
- Utilized Terraform and Ansible for Infrastructure as Code (IaC) to provision and manage AWS resources (EC2, S3, RDS, Lambda), optimizing deployment of scalable healthcare applications with 99.9% uptime.
- Wrote comprehensive unit, integration and end-to-end tests using PyTest, Selenium and Postman, achieving 95% code coverage and ensuring robust, error-free deployment of healthcare applications.

EDUCATION:

Master of Computer Science - PACE University, New York, USA

Bachelor of Electronics and Communication Engineering (ECE) – SRM University, Chennai, India